Chemistry logs

SOP:

1. Weigh a fixed amount of oxalic acid and add any excess weight of dust, including the stir bar in the flask.
2. Heat the oil bath on a hotplate to a defined temperature and speed.
3. Once the temperature is reached, add a defined amount of water to the flask to achieve the desired oxalic acid concentration (see bio section), and place the flask into the oil bath as far as possible.
4. Allow the leaching to occur for a defined time.
5. After the time has elapsed, remove the flask and filter it into another flask/glass vessel using a filter. The oil bath can be heated up at the same time (over 100 degrees Celsius, the temperature doesn't matter).
6. Then it depends on the amount of solution.
7. For a large amount of solution: set up a simple distillation temperature:
8. Place a reflux flask centrally behind the oil bath so that the flask with the filtered leaching solution can remain in the oil bath, with another flask on the other side to collect the distilled water.
9. Connect the reflux condenser to the water supply and, on the other side, connect a hose to the tank's drain.
10. When enough of the solution has been distilled (not all of it, if there's still some left), or immediately if the amount of solution is small:
11. Transfer to a small Schott flask.
12. Immerse the Schott flask as deeply as possible in the oil bath.
13. When no more liquid is visible, remove the Schott flask and dismantle everything.

Date	28.7.
Who	Daniel, Nathanael
What	Leaching experiment ○ Duration: 5 hours ○ Temperature: 60°C

Date	29.7.
Who	Daniel, Nathanael
What	Leaching experiment ○ Duration: 6 hours ○ Temperature: 60°C

Date	30.7.
Who	Daniel, Nathanael
What	Leaching experiment ○ Duration: 7 hours ○ Temperature: 60°C

Date	1.8.
Who	Daniel, Nathanael, Ayleen
What	Leaching experiment ○ Duration: 8 hours ○ Temperature: 60°C

Date	4.8.
Who	Daniel, Nathanael
What	Leaching experiment ○ Duration: 5 hours ○ Temperature: 50°C

Date	5.8.
Who	Daniel, Nathanael, Johannes
What	Leaching experiment ○ Duration: 5 hours ○ Temperature: 60°C

Date	6.8.
Who	Nathanael, Johannes
What	Leaching experiment ○ Duration: 5 hours ○ Temperature: 70°C

Date	7.8.
Who	Daniel, Nathanael, Johannes

What	Leaching experiment ○ Duration: 5 hours ○ Temperature: 80°C